XPL Layout and Aisle Planning Specification(Euro Pallets)

1. Traffic Aisle Width Requirements

Minimum Aisle Width Table

Action Type	Channel Type	Minimum Width
Driving	One-way road	1.5 m
Driving	Two-way road	3.0 m
Turning	One-way road	2.3 m
Turning	Two-way road	3.8 m
Loading / Unloading	One-way, drop-off on one side	3.0 m
Loading / Unloading	One-way, drop-off on both sides	3.5 m
Loading / Unloading	Two-way, drop-off on one side	4.0 m
Loading / Unloading	Two-way, drop-off on both sides	4.5 – 5.6 m

Note:

More aisle space allows:

- Higher driving speed
- Larger safety detection fields
- Faster pallet throughput
- Easier commissioning and tuning

2. Turning Area Requirements

Turning Rules

- During turning, the XPL automatically:
  - Reduces speed
  - Switches safety lidar to turning protection mode

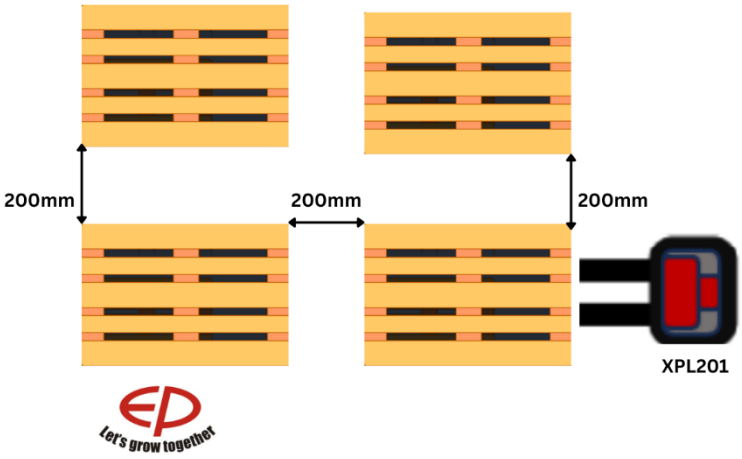
Practical Guidance

- Recommended minimum widths:
  - ≥ 2.3 m for one-way turning
  - ≥ 3.8 m for two-way turning

3. Storage Space Requirements

Pallet Spacing

- Recommended gap between pallets:
  200 – 300 mm



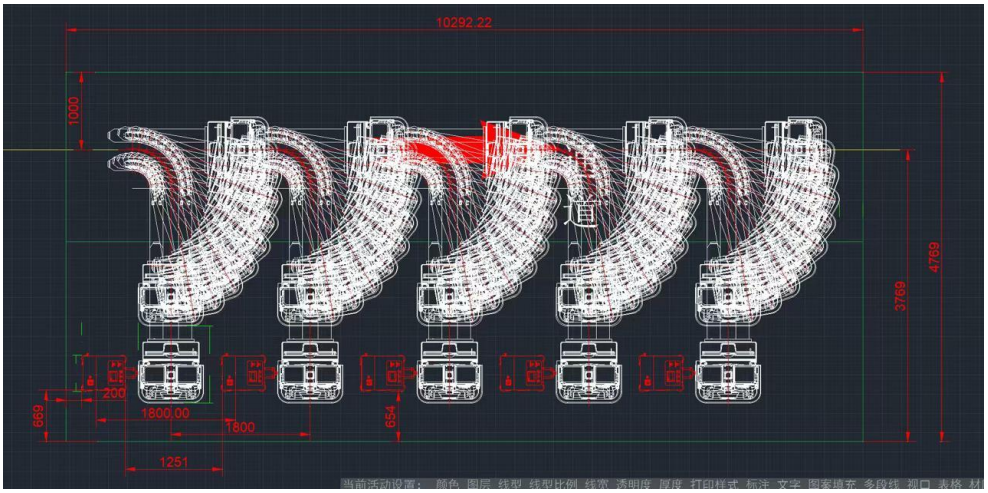
4. Speed vs Aisle Width

According to ISO 3691-4 safety principles, vehicle speed is limited by available safety distance, which is directly related to aisle width.

Typical Speed Guidance (XPL)

	Clearance between the truck and its load against the continuous fixed closed structure[d]		Clearance from the current position to the fixed closed structure/object in the direction of travel, C3	Personnel detection means in travel direction (PLd)	Max speed[e]	Required zone classification[a]	Reachable stop function required within 600 mm	Floor/ground marking or extra warnings required	Automatic restart permitted
	Clearance on one side, C1 mm	Clearance on the other side, C2 mm	mm						
1a	> 500	> 500	> 500	ACTIVE	Rated speed	Operating	NO	NO	YES
1b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
2a	> 500	> 500	< 500	ACTIVE[f]	0,7 m/s	Operating hazard	NO	YES	YES
2b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
3a	> 500	< 500 and > 100	> 500	ACTIVE	1,2 m/s	Operating hazard	NO	YES	YES
3b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
4a	> 500	< 500 and > 100	< 500	ACTIVE[f]	0,7 m/s	Operating hazard	NO	YES	YES
4b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
5a	> 500	< 100	> 500	ACTIVE	Rated speed	Operating	NO	NO	YES
5b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
6a	> 500	< 100	< 500	ACTIVE[f]	0,7 m/s	Operating hazard	NO	YES	YES
6b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
7a	< 500 and > 100	< 500 and > 100	> 500	ACTIVE	1,2 m/s	Operating hazard	NO	YES	NO
7b				MUTED	0,3 m/s	Restricted	NO	YES	—[b]
8a	< 500 and > 100	< 500 and > 100	< 500	ACTIVE[f]	0,3 m/s	Operating hazard	YES[c]	YES	NO
8b				MUTED	0,3 m/s	Restricted	YES[c]	YES	NO
9a	< 500 and > 100	< 100	> 500	ACTIVE	1,2 m/s	Operating hazard	NO	YES	NO
9b				MUTED	0,3 m/s	Restricted	NO	YES	—[b]
10a	< 500 and > 100	< 100	< 500	ACTIVE[f]	0,3 m/s	Operating hazard	YES[c]	YES	NO
10b				MUTED	0,3 m/s	Restricted	YES[c]	YES	NO
11a	< 100	< 100	> 500	ACTIVE	Rated speed	Operating hazard	NO	YES	YES
11b				MUTED	0,3 m/s	Operating hazard	NO	YES	—[b]
12a	< 100	< 100	< 500	ACTIVE[f]	0,3 m/s	Operating hazard	YES[c]	YES	NO
12b				MUTED	0,3 m/s	Restricted	YES[c]	YES	NO

5. Automatic Charging Station Layout

Recommended Space

Item	Requirement
Station length	3.6m
Station width	1.5m
Side clearance	≥ 300 mm (each side)
Rear clearance	≥ 500 mm
Overhead clearance	No reflective or glass surfaces